

Abhishri Ajit Medewar

(602) 621-2059 | abhishri.medewar@outlook.com | [Portfolio](#) | [Github](#) | [Linkedin](#) | [Medium](#)

EDUCATION

Master of Computer Science, Arizona State University
Relevant Coursework: Machine Learning, Deep Learning

August 2021 - May 2023
GPA 3.8

Bachelor of Computer Engineering, Pune University

Relevant Coursework: Data Structures, Database Management Systems, Software Engineering

August 2014 - May 2018
GPA 3.7

SKILLS

Programming Languages: Python, C++, C

Frameworks and Libraries: PyTorch, Tensorflow, Scikit-learn, Numpy, Scipy, Pandas, Matplotlib, Seaborn, Keras, OpenCV, BeautifulSoup, AWS, FFmpeg, Docker, S3, Hadoop, Kubernetes, MLflow, Weights & Biases, ONNX, TensorRT

Web Technology and Database: Flask, FastAPI, React, Angular, HTML, CSS, MySQL, PostgreSQL, MongoDB

Operating System/Developer Tools: Linux, Windows, Git, Jira, VS Code, Sublime text, Visual Studio, PyCharm

EXPERIENCE

Machine Learning Engineer, vivoVerse, Austin, USA

October 2023 - Present

- Developed and deployed a RetinaNet object detection model, leveraging mosaic augmentation technique resulting in **2% accuracy improvement**. Converted the **model into ONNX format** to achieve faster inference speed and ensure smooth deployment in production environment.
- Implemented and integrated **data pipelines into ML platform** to facilitate model training processes. Utilized **MLflow** for comprehensive testing, version control, and real-time monitoring of machine learning models.
- Implemented a vision **transformer** from scratch for classification of images with dimensions of 5000x3000, leading to a **5% enhancement** in classification accuracy.
- Developed streamlined procedures for the storage, retrieval, and manipulation of large scale(terabytes) datasets utilizing **PostgreSQL**, enhancing overall efficiency.

Machine Learning Engineer Intern, Intel Corporation, Santa Clara, USA

June 2022 - April 2023

- Implemented **distributed training** for ML model and performed hyperparameter optimization. This effort led to a remarkable **40% reduction** in training time.
- Worked on reproducing State-of-the-Art research paper results, leveraged **Docker** for efficient training.

Machine Learning Engineer, Tata Elxsi, Pune, India

July 2018 - June 2021

- Implemented State-of-the-Art region-based object detector using **Faster-RCNN** with **98%** accuracy and average inference time of 3sec/image on **Jetson TX2**, with image dimension of 2592x1944.
- Deployed an InceptionV2 and Faster-RCNN based object detector on **Nvidia Jetson TX2** using Tensorflow C++ API and **bazel**.
- Implemented **k-means clustering algorithm** for detecting small dust particles on camera lens, anti-reflection glass for optical system with overall accuracy of **95%**.

PROJECTS

Deep Background: U-Net-Based Image Background Segmentation and Editing [[Github](#)]

Summer 2023

- Implemented U-Net architecture from scratch, achieving an impressive 98.90% accuracy (dice score) in background segmentation using the Human Segmentation Dataset.
- Employed TensorRT to convert the trained model into an optimized TensorRT format, resulting in a significant 6x performance boost.

DeepLearn: Simplified CNN Training and Evaluation for Any Dataset [[Github](#)]

Spring 2023

- Implemented State-of-the-Art CNN architectures from scratch such as GoogLeNet / InceptionV1, AlexNet, VGG, LeNet, and ResNet with custom dataloader.
- Utilized **FastAPI** to develop RESTful APIs, enabling streamlined model deployment and seamless integration within application.

Deep Learning Approach for Human Stress Detection [[Github](#)]

Summer 2018

- Developed custom CNN for detection of stress level (0-9) by using FER2013 dataset to identify the facial emotion of the person from an image and correlating the emotion percentage with stress level with 85% accuracy.
- Utilized Flask to develop APIs for serving models, facilitating seamless deployment of model.

ACHIEVEMENTS

- Received **Tata Elxsi Bravo Award** for outstanding work in machine learning model fine-tuning.
- Exhibited strong leadership by driving Proof of Concept work at Tata Elxsi and securing funding from key clients.